

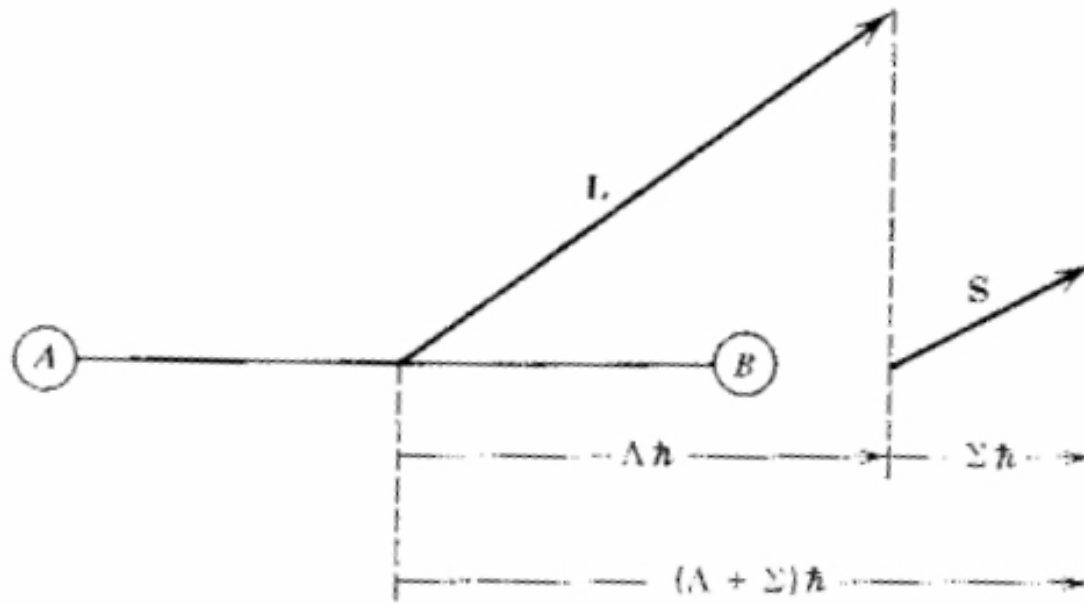
Diatomic Molecules Real Behavior

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Molecular Electronic States

- A diatomic molecule may assume different electronic states.
- For atoms, the electron wavefunctions are approximately spherically symmetric and L , the angular momentum, is quantized.
- Diatomic molecules are not spherically symmetric. Only the component of L along the inter nuclear axis is quantized.

Angular Momentum



Angular Momentum

- The component of L along the inter nuclear axis is

$$L_{AB} = \pm \Lambda \hbar, \quad \Lambda = 0, 1, 2, 3, \dots$$

- We use a Greek capital letter to designate Λ

Λ :	Σ	Π	Δ	Φ	Γ	...	
Symbol	=	0	1	2	3	4	...

Electron Spin and Spin-Orbit Coupling

- Like orbital angular momentum, spin is quantized along the internuclear axis.

$$S_{AB} = \pm \Sigma \hbar, \quad \Sigma = -S, -S + 1, \dots, S - 1, S$$

- As with atomic systems, the multiplet structure is a consequence of spin-orbit coupling.
- If T_0 is the term energy without coupling, then

$$T_e = T_0 + C \Lambda \Sigma$$

$$C = 1 - 10^3 \text{ cm}^{-1} \quad (T_0 = 10^4 - 10^5 \text{ cm}^{-1})$$

Coupling constant

Multiplet Components

Λ	Number of Components
0	1
1,2,3,...	$2S+1$

Spectroscopic Term Classification

- Like atoms we classify the electronic energy levels of a diatomic molecule by a term symbol of the form

$$^{2S+1}\Lambda_{\Lambda+\Sigma}$$

- $2S+1$ is the multiplicity (although there is no splitting if $\Lambda = 0$.)

Symmetry Properties

1. For a molecule with identical nuclei, if upon reflection of the electron coordinates the electronic amplitude remains unchanged, a right subscript g (German gerade, meaning “even”) is used in the term classification. If the amplitude function changes sign, a u (German ungerade, meaning “odd”) is used

$$\Sigma_g, \Pi_g, \dots \quad \Sigma_u, \Pi_u, \dots$$

Symmetry Properties

2. Depending on whether the electronic amplitude changes sign with reflection in a plane through the internuclear axis, Σ states ($\Lambda = 0$) are designated as

Σ^+ (no change) or Σ^- (change)

Structure of the Diatomic Molecule

- Energy

$$T_{tot} = T_0 + G(v) + F_v(J)$$

- Degeneracy

$$\Lambda = 0 \quad g_e = 2S+1$$

$$\Lambda > 0 \quad \text{With spin-orbit splitting } g_e = 2$$

$$\Lambda > 0 \quad \text{No spin-orbit splitting } g_e = 2(2S+1)$$

$$- \quad g_v = 1$$

$$- \quad g_J = 2J+1$$

OH Energy Level Diagram

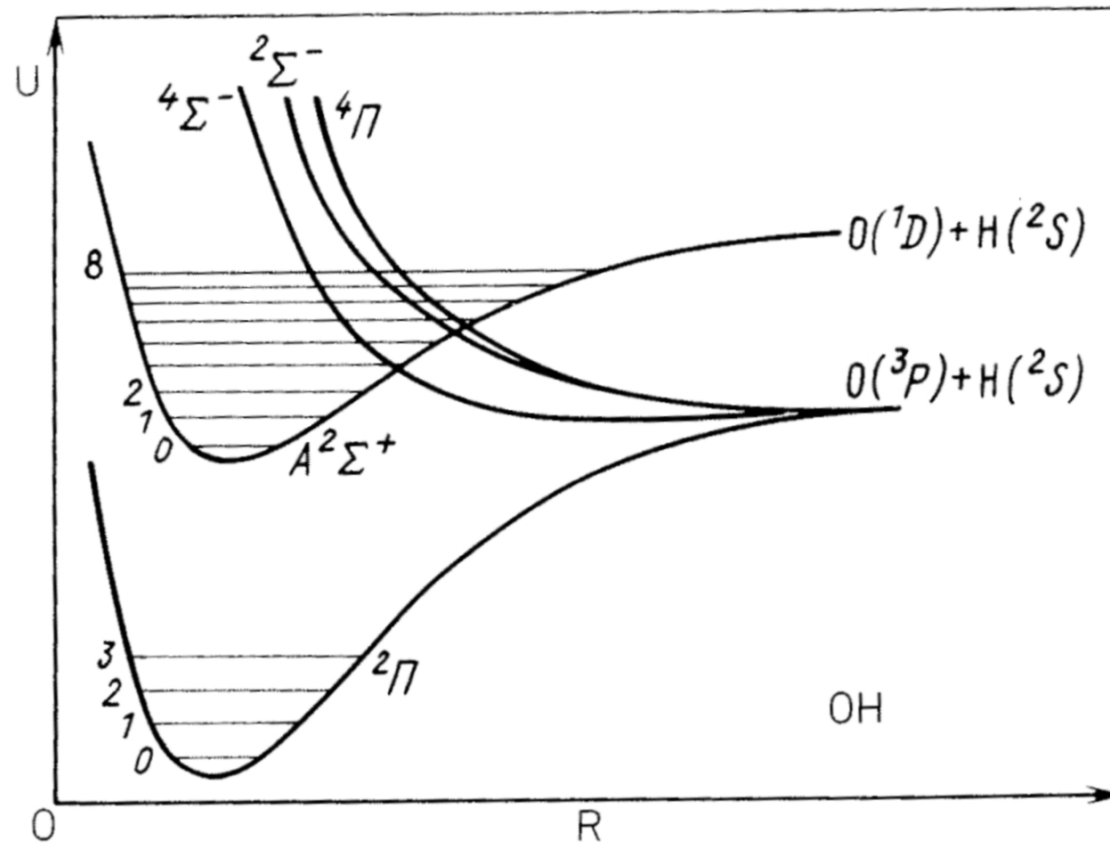


Fig. 10.12. Qualitative scheme of OH adiabatic terms

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Diatomic constants for ¹⁸O⁺H

State	T _e	ω _e	ω _e x _e	ω _e y _e	B _e	α _e	γ _e	D _e	β _e	r _e	Trans.	v ₀₀
Theoretical potential functions for 48 states Easson and Pryce, 1973 ; for X ² Π and A ² Σ ⁺ see Stevens, Das, et al., 1974 Chu, Yoshimine, et al., 1974 Meyer and Rosmus, 1975 .												
C ² Σ ⁺	89459. ₁	1232.9 Z	19.1		4.247 1 2	0.078		2E-4		2.046 ₁	C → A 3 4 R	55820.7 Z
	↳ Michel, 1957 ; Felenbok, 1963 ; <i>missing citation</i>											
C ² Σ ⁺ 5											(C → X) 3	(88223)
	↳ Michel, 1957 ; <i>missing citation</i>											
D ² Σ ⁻	(82130)	(2954)			[15.2179] 6			[16.16E-4]		[1.0809 ₃]	D → X 7 R	81759.7 ₈ 8 Z
	↳ Douglas, 1974											
State	T _e	ω _e	ω _e x _e	ω _e y _e	B _e	α _e	γ _e	D _e	β _e	r _e	Trans.	v ₀₀
B ² Σ ⁺	69774	[660.0] Z	9		[5.086] 10 2	11		[9.29E-4] 12		[1.869 ₈]	B → A 4 R	35965.5 Z
	↳ Barrow, 1956 ; Michel, 1957 ; Herman, Felenbok, et al., 1961 ; <i>missing citation</i> ; <i>missing citation</i> ; <i>missing citation</i> ; Czarny, Felenbok, et al., 1971											
A ² Σ ⁺	32684.1	3178.8 ₆ Z	92.91 ₇ 13		17.358 14 15 16 2	0.786 ₈ 17	-0.016	[20.39E-4] 18		1.0121	A ↔ X 19 20 R	32402.3 ₉ 8 Z
	↳ Dieke and Crosswhite, 1948 ; Stoebner and Delbourgo, 1967 ; <i>missing citation</i>											
X ² Π ₁	0 21	3737.76 ₁ Z	84.881 ₃ 22		18.910 ₈ 23 2	0.7242 24		19.38E-4 25		0.96966	1/2 + 3/2 26	126.23
	↳ Evenson, Wells, et al., 1970 ; Mizushima, 1972											
	Rotational-vibration band 27 28											
	1-0 sequence											
	↳ Allen, Balne, et al., 1957 ; Charters and Polanyi, 1960 ; Maillard, Chauville, et al., 1976											
	2-0 sequence											
	↳ Benedict, Plyler, et al., 1953 ; Charters and Polanyi, 1960 ; Rogge, Yarger, et al., 1960 ; Maillard, Chauville, et al., 1976											
	3-0 sequence											
	↳ Bass and Garvin, 1962 ; Maillard, Chauville, et al., 1976											
	4-0 sequence											
	↳ Herman and Hornbeck, 1953 ; Chamberlain and Roesler, 1955 ; McKinley, Garvin, et al., 1955 ; Bass and Garvin, 1962											
	other sequences											
	↳ Chamberlain and Roesler, 1955 ; McKinley, Garvin, et al., 1955 ; Blackwell, Ingham, et al., 1960 ; Bass and Garvin, 1962											
	Rotation sp.											
↳ Madden and Benedict, 1955 ; Ducas and Javan, 1974 ; Downey, Robinson, et al., 1977												
Hf A-doubling sp. 29 30 31												
↳ Dousmanis, Sanders, et al., 1955 ; Ehrenstein, 1963 ; Poynter and Beaudet, 1968 ; Destombes, Marliere, et al., 1974 ; Destombes and Marliere, 1975 ; Meerts and Dymanus, 1975 ; Meerts, 1977												
EPR sp. 32												
↳ Radford, 1961 ; Churg and Levy, 1970 ; Clough, Curran, et al., 1971 ; Lee, Tam, et al., 1971 ; Hinkley, Walker, et al., 1973 ; Lee and Tam, 1974												

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